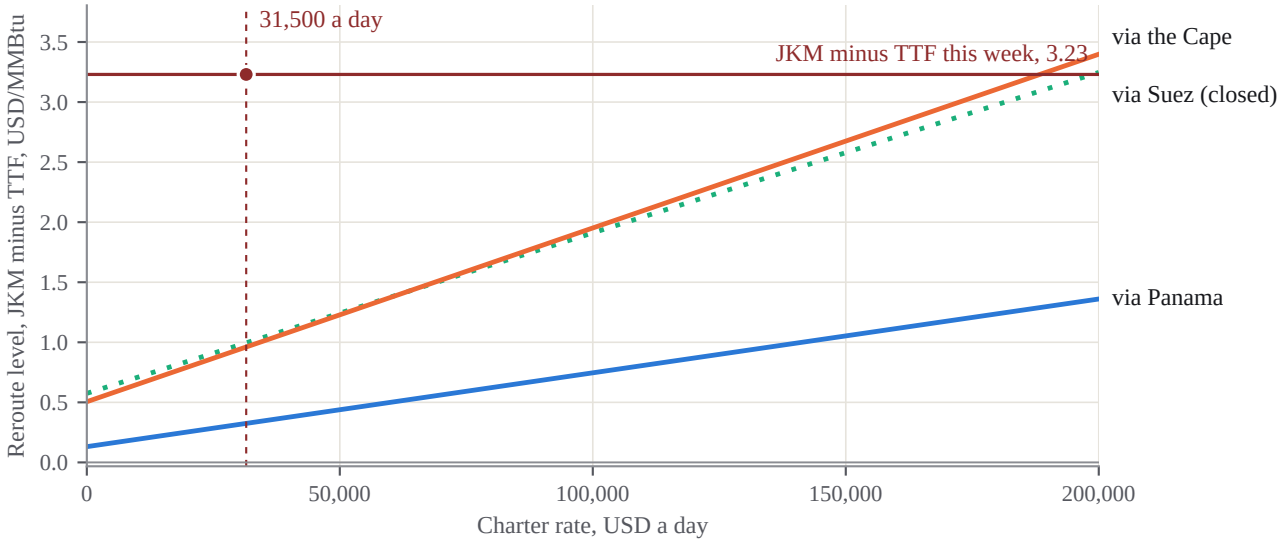
# When a US Gulf LNG cargo pays to go east

**JKM is 3.23 USD/MMBtu above TTF. At 31,500 USD a day, the last charter rate reported, on 2 October 2026, a cargo loaded at Sabine Pass pays to go to Japan instead of Rotterdam above a spread of 0.32 via Panama and 0.96 via the Cape, so the arb is open on both routes.**

A cargo delivered to Northwest Europe sells 0.76 under TTF; without that discount the levels would be 1.10 and 1.75 (ACER, to 27 October 2026). Each 10,000 USD a day on the charter rate adds 0.06 to the Panama level and 0.14 to the Cape level, because the Cape adds 53 days to the round trip and Panama 23. With the ship already paid for the levels are 0.13 and 0.51.

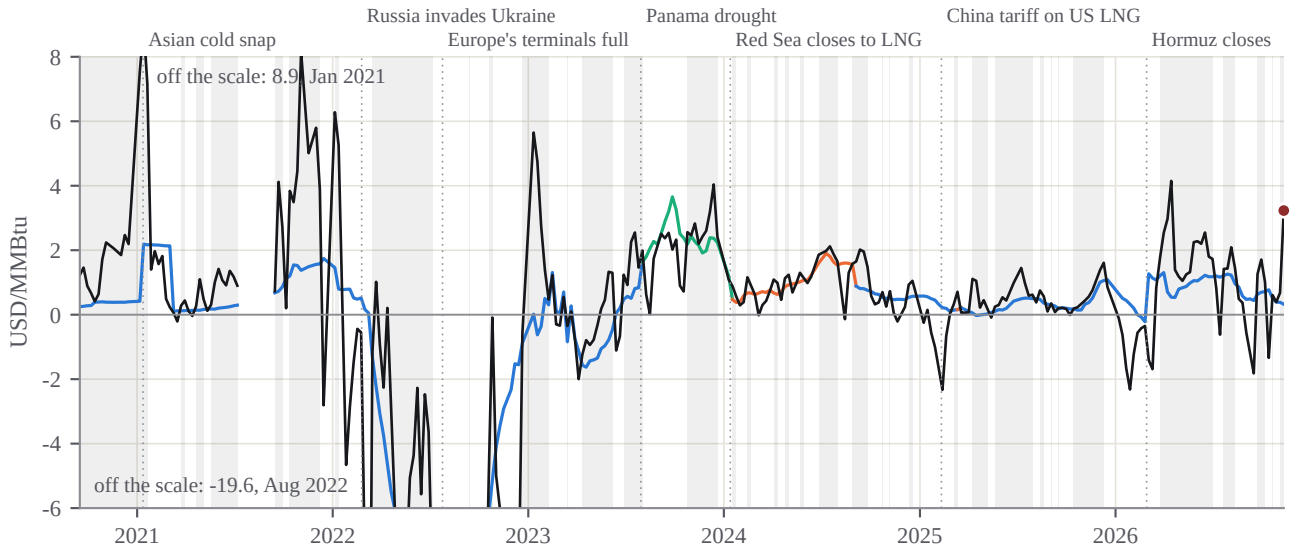
**Chart 1. The longer the route, the faster its reroute level rises with the charter rate**




| *Reroute level by charter rate, USD/MMBtu* | *ship paid for, 0* | *low, 26,500* | *central, 43,250* | *high, 64,750* | *arb closes above* |
|---|---|---|---|---|---|
| via Panama | 0.13 | 0.29 | 0.40 | 0.53 | 504,130 USD a day |
| via Suez (closed) | 0.57 | 0.93 | 1.15 | 1.44 | 198,732 USD a day |
| via the Cape | 0.51 | 0.89 | 1.13 | 1.44 | 188,361 USD a day |

**Chart 2. The spread cleared the level in 168 of 289 weeks: among full years, most often in 2021 and least in 2022**

JKM minus TTF (black) and the reroute level of the cheapest open route at the reported charter rate (coloured by route). Shaded: the arb is open. With the ship already paid for it was open in 226 weeks.

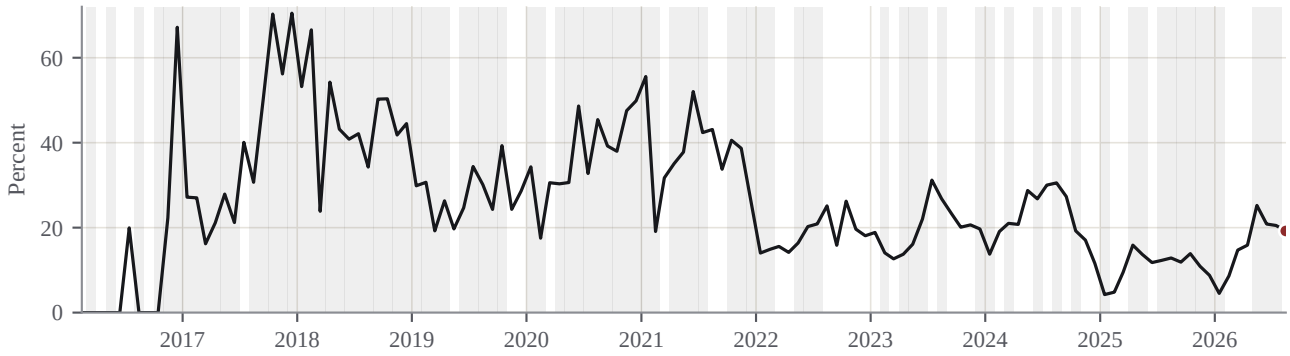


## Did cargoes follow?

After a month with the arb open, 29 percent of US LNG left for Japan, South Korea, China and Taiwan; after a closed month, 19 percent. The gap belongs to the years before 2022, when the arb was open in 60 months of 65 and the share was 35 percent; since 2022 it is 17 percent after an open month and 19 after a closed one. Europe's pull, term contracts and China's tariff on US LNG have kept Asia's share low whatever the arb (120 months of the Department of Energy's cargo record; the rule was fixed before the result was computed).

**Chart 3. Asia's share of US LNG halved after 2021, and within neither period does it follow the arb**

Share of US LNG sent to JKM markets, by departure month. Shaded: the arb was open the month before.




## The same cargo answers two more questions

When does the arb close on the cargo itself? A buyer lifts a cargo when its best netback covers 115 percent of Henry Hub. From February to August 2020 a cargo sold in Europe did not, and about 171 US cargoes were cancelled.

What is a Panama slot worth? Through Panama the round trip is 30 days shorter than round the Cape. At this week's charter rate and gas price a ship can pay 2.63 million USD above the tariff for its two transits, or wait 9.5 days at each, before the Cape is the cheaper way east.

## Method

```
loaded     = capacity x fill x MMBtu per m3          174,000 m3 x 98.5 percent x 23 = 3,941,970 MMBtu
delivered  = loaded - 0.085 percent a day x round trip days       the boil-off is the ship's fuel
cost       = hire x round trip days + ports + canal tolls and fees + EU ETS (Europe) + financing
netback    = (DES price x delivered - cost) / loaded      Europe: TTF less the DES discount. Asia: JKM
S*(route)  = the JKM minus TTF spread at which netback(Asia, route) = netback(Europe)
H*(route)  = the charter rate at which the two netbacks are equal at today's prices
lift       = best netback - 115 percent of Henry Hub      the fixed liquefaction fee is sunk
```

Both prices are delivered ex-ship: Europe's is TTF less a discount this study measures from ACER's assessment, the charge for regasification and entry. Hire runs for the round trip, as in the Spark30S freight assessment, whose worked example the model reproduces to the dollar. This week: Gate 26.4 days and a netback of 26.44; Panama 49.6 days and 29.23; the Cape 79.6 days and 28.56, in USD per MMBtu loaded.

## Caveats

• The charter rate is one published assessment (Spark30S Atlantic), reported on 2 October 2026 and held since; a real fixture differs.
• Europe's DES discount is this study's measure from ACER's assessment, not ACER's own spread; until January 2023 it is an assumption.
• Panama is priced at the tariff with a booked slot. Slots are auctioned and their price is not published.
• Prices are weekly averages of front months, which can name different delivery months; a cargo is sold further out.
• Flows show association, not cause. Nothing here is a forecast or a trading signal.

Sources: U.S. Energy Information Administration (JKM and TTF, credited to Bloomberg Finance L.P.; Henry Hub), ACER, U.S. Department of Energy, Spark Commodities as reported by LNG Prime, Panama Canal Authority, Suez Canal Authority, Federal Reserve Board, Federal Reserve Bank of New York, European Commission and DEHSt, METI, World Bank. Every figure and its source: the site's Method view.